Literature answer to the property-[H] characterization

arXiv:0801.0085, Section 2 $\longrightarrow$ arXiv:0812.0889, Theorem 2.5

Status: literature_already_answered.

Original source. Jacek Chmieliński, Dijana Ilišević, Mohammad Sal Moslehian, and Ghadir Sadeghi, *Perturbation of the Wigner equation in inner product C^*-modules*, arXiv:0801.0085; J. Math. Phys. **49** (2008), 033519. On PDF page 4, immediately after defining condition [H], the authors write:

> It is an interesting question to characterize the class of all inner product C^*-modules in which [H] is satisfied.

Here [H] says that every norm-bounded sequence (x_n) has a subsequence (x_{n_k}) and x_0 such that $\|\langle x_{n_k}, y\rangle - \langle x_0, y\rangle\| \to 0$ for every y.

Decisive later answer. Ljiljana Arambašić, Damir Bakić, and Mohammad Sal Moslehian, *A characterization of Hilbert C^*-modules over finite dimensional C^*-algebras*, arXiv:0812.0889; Operators and Matrices **3** (2009), 357–364. Its abstract explicitly says that it answers the question posed in the original paper. Theorem 2.5 on PDF page 5 states:

> A full Hilbert C^*-module over a C^*-algebra A satisfies [H] if and only if A is finite dimensional.

Identification and full scope. For an arbitrary Hilbert A-module X, let

$$J = \overline{\operatorname{span}}\{\langle x, y\rangle : x, y \in X\}.$$

Then J is the coefficient ideal seen by X, and the same inner product and norm make X a full Hilbert J-module. Condition [H] is unchanged. Thus the characterization requested by the source is equivalently

$$\boxed{X \text{ satisfies [H]} \quad \Longleftrightarrow \quad J \text{ is finite dimensional.}}$$

The reverse implication is the original paper's Proposition 2.1; the later paper proves the converse for full modules and explicitly presents it as the answer to the source question.

Limitations. This status closes only the property-[H] characterization. It does not claim to settle the source paper's separate critical-control problem $\varphi(x, y) = \varepsilon\|x\|\|y\|$ or its closing uniqueness question for exact Wigner approximants. Several direct counterexample attempts for the latter are recorded in the linked attempt note but were not promoted.

Evidence and verification. Searches by the exact title, question phrase, property [H], and Hilbert C^*-module located arXiv:0812.0889. The source question and supporting theorem were checked in the official PDFs saved with this note. The supporting authors explicitly cite the original paper and say that their theorem answers its question.